ARIS

Automated Regulatory Intelligence System

User Manual

Version 1.2 · March 2026

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About This Manual

This manual covers installation, configuration, daily operation, and advanced use of ARIS. Version 1.2 adds documentation for the Windows UTF-8 encoding fix required for Python files containing non-ASCII characters.

Part	Contents
Part 1 - Installation	System requirements, installation, Windows setup notes
Part 2 - First Run	Getting started, first-run behaviour, dashboard tour
Part 3 - Core Workflows	Daily use, gap analysis, comparison, synthesis, horizon
Part 4 - Configuration	API keys, jurisdictions, scheduling, notifications, LLM providers
Part 5 - Reference	All views, CLI commands, troubleshooting, glossary

Part 1 - Installation

System Requirements

Requirement	Details
Python	3.11 or later (required)
Node.js	18 or later (required for Word export)
Operating system	macOS 12+, Ubuntu 20.04+, or Windows 10+ (WSL2 recommended)
RAM	4 GB minimum, 8 GB recommended
Disk	1 GB for application + variable for document database
Internet	Required for fetching from government APIs; ARIS itself runs locally

Step-by-Step Installation

1. Clone or download the project

```
git clone <your-repo-url>
cd ai-reg-tracker
```

2. Install Python dependencies

```
pip install -r requirements.txt
pip install pdfplumber pypdf   # optional - for PDF extraction
```

3. Install Node.js dependencies (for Word export)

```
npm install -g docx
```

4. Configure API keys

```
cp config/keys.env.example config/keys.env
```

Edit config/keys.env and fill in your API keys. At minimum you need ANTHROPIC_API_KEY.

5. Run the database migration

```
python migrate.py
```

Creates all 17 database tables. Safe to run multiple times.

6. Build the browser interface

```
cd ui && npm install && npm run build && cd ..
```

7. Start the server

```
python server.py
```

Open http://localhost:8000 in your browser.

Windows Setup Notes

ARIS is developed on Unix/macOS but runs on Windows. Two issues are known on Windows:

SyntaxError: Non-UTF-8 code

Python on Windows defaults to ASCII for source files. Any file containing non-ASCII characters (em dashes in comments, box-drawing characters in section dividers) will fail to import with:

```
SyntaxError: Non-UTF-8 code starting with '\x97' on line 3
```

All ARIS Python files include # -*- coding: utf-8 -*- as their first line to declare UTF-8 explicitly. If you receive this error after updating files, run this from your project root to restore the declarations:

```
import pathlib
DECL = b'# -*- coding: utf-8 -*-\n'
for p in pathlib.Path('.').rglob('*.py'):
    if 'node_modules' in str(p) or '__pycache__' in str(p): continue
    raw = p.read_bytes()
    if b'coding: utf-8' not in raw[:50]:
        p.write_bytes(DECL + raw)
        print(f'fixed: {p}')
```

Then restart python server.py.

npm run build fails with "stream did not contain valid UTF-8"

Same issue in .js or .jsx files. Open the affected file (usually vite.config.js or a file in ui/src/) in a text editor and save it as UTF-8 encoding. In Notepad: File > Save As > Encoding dropdown > UTF-8.

Security: Network Binding

The server binds to 127.0.0.1 (localhost only) by default. ARIS has no authentication layer.

To allow access from other devices on your local network, add to config/keys.env:

```
ARIS_HOST=0.0.0.0
```

> ⚠ **Setting ARIS_HOST=0.0.0.0 exposes the full API to anyone on your network. Use only on trusted networks - no login or access control exists.**

After Any Update

```
python migrate.py          # adds any new tables or columns safely
cd ui && npm run build     # only if UI source files changed
python server.py           # restart the server
```

Part 2 - First Run

What Happens on First Run

- The server indexes all 31 baseline regulations into the RAG index (~5 seconds, background thread). Ask ARIS answers baseline questions immediately.
- The dashboard shows a setup prompt guiding you through the first fetch.
- No API keys other than ANTHROPIC_API_KEY are strictly required for a first run.

Running Your First Fetch

Navigate to Monitor > Run Agents. A yellow banner will appear explaining that Force Summarize is enabled automatically for the first run so all documents process fully without the pre-filter blocking unlearned sources.

Click Run All Sources. A full run takes 2-3 minutes with all source tracks running concurrently.

Post-Run Summary Card

Row	Meaning
Fetched N new documents	Documents added or updated across all enabled sources
Summarised N	Documents processed by Claude
Skipped N (pre-filter)	Filtered before Claude - visible in Documents view with exact reason and score
Auto-archived N	Documents Claude rated <= 0.15 relevance - moved to Archive automatically
Changes N (M critical)	Version diffs and addenda detected against baselines
Urgency pills	Distribution of Critical / High / Medium / Low across summarised docs

"Skipped" documents are not lost. They appear in Documents view with a grey badge showing the filter reason. Use Force Summarize to reprocess them.

The Dashboard

Alert Rail

Renders only when something needs attention: unreviewed Critical changes, horizon deadlines within 30 days, Critical-urgency documents, or a large pending backlog.

Insight Grid

Regulatory Pulse (velocity sparklines per jurisdiction) and What's Active (coverage stats, upcoming deadlines).

Horizon Widget

Full-width panel showing upcoming regulatory deadlines with four urgency buckets (Within 30d / 30-180d / 180d+ / TBD), scrollable item list with stage badges, jurisdiction chips, and day countdowns. Always populated - 17 confirmed upcoming events are seeded.

System Health

Data coverage %, domain split, baselines loaded, API key status. Red border if Anthropic key is missing.

Part 3 - Core Workflows

Domain Filter

Every data view has a three-pill domain filter: All / AI Regulation / Data Privacy. Each view remembers its own setting independently.

Domain	What it covers
AI Regulation	Risk classification, transparency obligations, prohibited uses, human oversight, conformity assessment, foundation models, automated decision-making rules
Data Privacy	Consent, individual rights, breach notification, legal bases, data transfers, DPO requirements, sector-specific privacy rules
Both	Documents covering both domains, or when you want the full picture

Reviewing Documents

Navigate to Monitor > Documents. Click any document to open the detail panel where you can read the plain-English summary, mark relevance, and review action items and deadlines.

Marking Not Relevant removes the document from the active list and moves it to Archive. The feedback updates source quality scores and keyword weights for future runs.

> ⚠ **Skipped documents appear with a grey badge and the exact filter reason and score. Use Run Agents > Force Summarize to reprocess them.**

Autonomous Learning

Skipped stubs feed the learner

Every processed document - whether it received a real summary or was filtered at any stage - calls record_auto_feedback() with the relevance score. Pre-filter rejections send a strong negative signal (score 0.0). Over time, sources consistently producing irrelevant documents are down-weighted automatically.

Auto-archive

Documents where Claude rates relevance <= 0.15 are automatically marked not_relevant (stamped user="aris_auto") and move to the Archive tab immediately.

Domain-keyed profiles

AI and privacy sources accrue quality scores independently. Marking AI documents as irrelevant does not affect privacy source scoring.

False-Positive Protection

- Stage 1: Unambiguous AI terms pass immediately ("artificial intelligence", "automated decision", "deepfake")

- Stage 2: Known false-positive patterns blocked (NAIC, MAID, PAID leave, BRAIN Initiative) unless 2+ additional AI terms present
- Stage 3: Scored match against 150+ term taxonomy; ambiguous terms (e.g. "aida") only count when backed by at least one unambiguous AI term

Relevance validation runs against the bill title only, not title + search keyword. The old approach caused every bill in a LegiScan result set to pass automatically.

Change Detection

Severity	Meaning
Critical	New prohibition, new mandatory obligation, or penalty change vs the baseline
High	Scope expansion, new requirements, or significant deadline change
Medium	Modified requirements, clarification, or amended scope
Low	Procedural change, minor amendment, or formatting update

Gap Analysis

Creating a company profile

- Click "New Profile" - fill in company name, jurisdictions, AI systems deployed, and industry
- Be specific about AI use cases - Claude uses this to identify which obligations actually apply

Running a gap analysis

- Select a profile and click "Run Gap Analysis"
- ARIS runs two Claude passes: scope mapping (which regulations apply), then gap identification (what you are missing)
- Results: posture score (0-100), gap breakdown by severity, compliant areas, phased roadmap

Exporting

Click "Export .docx" to download a formatted Word document. Requires Node.js and the docx npm package.

Regulatory Horizon

Navigate to Monitor > Horizon for the full forward-looking calendar. Sorted by anticipated date. Dismiss items not relevant to your organisation.

Date	Jurisdiction	Event
Jun 30, 2026	Colorado	Colorado AI Act - full compliance required (SB 24-205)
Aug 2, 2026	EU	EU AI Act - GPAI model obligations apply

Aug 2, 2026	California	California AI Transparency Act (SB 942) effective
Sep 12, 2026	EU	EU Data Act - data sharing obligations fully applicable
Oct 2026	UK	UK DPDI Act - secondary legislation / ICO AI codes
Nov 2026	EU	EU eIDAS 2.0 - digital identity wallet deployment
Aug 2, 2027	EU	EU AI Act - high-risk AI system obligations apply

Jurisdiction Comparison

Navigate to Research > Compare. Select any two of the 31 baselines and optionally enter a focus topic. Six suggested comparison pairs appear on the placeholder screen.

Synthesis

Navigate to Research > Synthesis. Enter a topic and optionally restrict to specific jurisdictions. Export as .docx.

Ask ARIS

Navigate to Research > Ask ARIS. The baseline index builds automatically at server startup. Example questions:

- What does GDPR require for breach notification?
- What are the prohibited uses of AI under the EU AI Act?
- What obligations does the Colorado AI Act impose on deployers?

Learning System

Navigate to System > Learning to review source quality profiles, keyword weight distributions, and active prompt adaptations. Toggle any adaptation off if it is being too aggressive.

Human feedback (Relevant / Partially Relevant / Not Relevant) updates source quality scores, keyword weights, and prompts. Autonomous learning additionally updates weights from every Claude relevance score above 0.7 (positive) or below 0.2 (negative).

Part 4 - Configuration

API Keys

LLM Provider

Setting	Description
LLM_PROVIDER	anthropic | openai | ollama | gemini (default: anthropic)
LLM_MODEL	Model name. Leave blank for provider default.
LLM_BASE_URL	For OpenAI-compatible endpoints (Groq, Together AI, Mistral, LM Studio).
ANTHROPIC_API_KEY	Required if LLM_PROVIDER=anthropic
OPENAI_API_KEY	Required if LLM_PROVIDER=openai (also used for Groq, Together AI, Mistral)
GEMINI_API_KEY	Required if LLM_PROVIDER=gemini

Ollama runs locally with no API key. Install from ollama.com, then: ollama pull llama3.1

Data Source Keys (all free)

Key	Source
REGULATIONS_GOV_KEY	Federal rulemaking dockets - open.gsa.gov/api/regulationsgov
CONGRESS_GOV_KEY	US Congressional bills and hearings - api.congress.gov/sign-up
LEGISCAN_KEY	All 18 US state legislatures, 30,000 calls/month - legiscan.com/legiscan
COURTLISTENER_KEY	Federal court opinions (optional) - free.law/recap/api

Security and System Settings

Setting	Default / Description
ARIS_HOST	127.0.0.1 - binds server to localhost only. Set to 0.0.0.0 for LAN access (no auth exists).
DB_PATH	./output/aris.db - SQLite database location
LOG_LEVEL	INFO - set to DEBUG for verbose output when troubleshooting

Domain Configuration

Setting	Default / Description
ACTIVE_DOMAINS	both - ai | privacy | both. Which regulatory domains to fetch.

LOOKBACK_DAYS	30 - days back for document searches on each run
MIN_RELEVANCE_SCORE	0.5 - minimum Claude relevance score to store a summary. Gate 3 threshold is 0.35.

Jurisdictions

Edit config/jurisdictions.py to control which states and international jurisdictions run on each fetch. Changes take effect on the next server restart.

Scheduled Monitoring

Option	Description
Enabled toggle	Turn scheduled runs on or off
Run every	6 / 12 / 24 / 48 hours
Domain	AI Regulation / Data Privacy / Both
Lookback	3 / 7 / 14 days - how far back to search on each scheduled run

Notifications

Email (SMTP)

```
NOTIFY_EMAIL=recipient@yourcompany.com
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=sender@gmail.com
SMTP_PASSWORD=your-app-password
```

For Gmail, generate an App Password at myaccount.google.com/apppasswords (requires 2FA enabled).

Slack webhook

```
SLACK_WEBHOOK_URL=https://hooks.slack.com/services/T.../B.../...
```

Resetting the Database

```
python reset.py                              # interactive menu (at project root)
python reset.py --documents                 # clear docs and summaries only
python reset.py --documents --learning       # clear docs + learning state
python reset.py --full --yes                 # clear everything except
profiles/gaps
```

Part 5 - Reference

All Browser Views

Section	View	Purpose
Monitor	Dashboard	Alert rail, pulse sparklines, horizon widget, system health
Monitor	Documents	Active documents with domain filter, urgency, Skipped badge with reason
Monitor	Changes	Version diffs with severity; side-by-side comparison
Monitor	Horizon	Forward-looking regulatory calendar; deadline countdowns; dismiss
Monitor	Watchlist	Keyword-based alerts with domain filter
Monitor	Enforcement	FTC / SEC / CFPB / ICO enforcement actions
Research	Baselines	Browse 31 baseline regulations; obligations, prohibitions, penalties
Research	Compare	Side-by-side Claude analysis of any two baselines
Research	Ask ARIS	RAG Q&A across all documents and baselines with citations
Research	Briefs	One-page jurisdiction briefs
Research	Synthesis	Cross-document narratives; conflict detection; .docx export
Research	Gap Analysis	Company-profile compliance gaps; roadmap; .docx export
Research	Obligations	De-duplicated obligation register across selected jurisdictions
Research	Timeline	Chronological regulatory timeline
Research	Concept Map	Cross-jurisdiction concept analysis
Research	Graph	Document relationship network
System	Run Agents	Trigger fetch/summarise; first-run banner; post-run summary with auto-archived count
System	PDF Ingest	Upload PDFs or trigger auto-download
System	Learning	Source quality profiles; keyword weights; prompt adaptations; toggle
System	Settings	API keys; jurisdictions; schedule; notifications; ARIS_HOST

CLI Reference

Command	Description
python main.py run	Full pipeline: fetch all sources, then summarise
python main.py run --domain ai	AI regulation only
python main.py run --domain privacy	Data privacy only

python main.py run --days 90	Longer lookback window
python main.py fetch --source federal	Federal sources only
python main.py fetch --source states	All enabled state legislatures
python main.py summarize	Summarise all pending documents
python main.py summarize --force	Bypass pre-filter (processes Skipped docs)
python main.py report	Print dashboard to terminal
python main.py changes --severity Critical	Show Critical changes only
python main.py baselines	List all 31 baselines
python main.py synthesise "topic"	Run cross-document synthesis
python main.py watch --interval 24	Run every N hours continuously
python main.py status	Show database statistics
python migrate.py	Run database migration (safe to repeat)
python reset.py	Interactive data reset tool
python reset.py --documents --learning	Clear docs + all learned weights and profiles
python reset.py --full --yes	Clear all except company profiles and gap analyses

Urgency Levels

Level	Meaning
Critical	Binding law with immediate or imminent compliance obligations or significant penalties
High	Enacted regulation with near-term deadlines, or proposed rule very likely to be finalised
Medium	Proposed or advancing regulation, or enacted rule with long compliance runway
Low	Guidance, voluntary frameworks, or regulatory activity unlikely to impose direct obligations
Skipped	Filtered before or after Claude. Reason and score visible in Documents view. Feeds learning agent.

Troubleshooting

SyntaxError: Non-UTF-8 code (Windows)

Python on Windows defaults to ASCII. Files with non-ASCII characters fail to import. All ARIS Python files include # -*- coding: utf-8 -*- as their first line. If you see this error after updating files, restore the declarations:

```
import pathlib
DECL = b'# -*- coding: utf-8 -*-\n'
for p in pathlib.Path('.').rglob('*.py'):
    if 'node_modules' in str(p) or '__pycache__' in str(p): continue
    raw = p.read_bytes()
    if b'coding: utf-8' not in raw[:50]:
        p.write_bytes(DECL + raw); print(f'fixed: {p}')
```

Then restart python server.py.

npm run build fails: "stream did not contain valid UTF-8"

Same encoding issue in a .js or .jsx file. Open the affected file in a text editor and save as UTF-8. In Notepad: File > Save As > Encoding dropdown > UTF-8. Most commonly affects vite.config.js.

Build fails: "Identifier STAGE_COLORS has already been declared"

Dashboard.jsx has duplicate const declarations. Rename the PulsePanel versions to PULSE_STAGE_COLORS and PULSE_JUR_COLORS. Replace Dashboard.jsx with the latest version from the repository.

"InterpreterAgent object has no attribute _write_skipped_stub"

_write_skipped_stub is a module-level function, not a method. Replace agents/interpreter.py with the latest version. The function definition must be present at module scope before the InterpreterAgent class.

Too many false positives from LegiScan

- Ensure state_agent_base.py validates relevance against the bill title only, not title + search keyword
- Ensure utils/search.py has the _NOT_AI_CONTEXTS blocklist and _STRONG_AI_SIGNALS fast path
- Run python reset.py --documents --learning --yes and let the system retrain from a clean slate

All documents Skipped on first run

Normal on a fresh install. Use Force Summarize for the first batch to build up feedback, then let normal filtering resume.

Export .docx fails

```
node --version           # must be 18+
npm list -g docx         # should show docx@9.x.x
npm install -g docx      # install if missing
```

Server not accessible from other devices

```
# Add to config/keys.env:
ARIS_HOST=0.0.0.0
```

Then restart. No authentication exists - only use on trusted networks.

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- You may use, copy, and modify ARIS for personal, educational, and non-commercial purposes
- You may not sell ARIS, offer it as a commercial product or service, or use it as the basis of a paid product without express written permission
- You may not provide ARIS to third parties as a hosted or managed service

The footer of every page in the ARIS browser interface displays:

> *"Not legal, compliance, or regulatory advice. ARIS is an informational research tool only. Nothing in this system constitutes legal, compliance, or regulatory advice. Always consult qualified legal counsel before making compliance or regulatory decisions."*

Glossary

Term	Definition
Auto-archive	Autonomous process that marks documents not_relevant (user="aris_auto") when Claude rates relevance <= 0.15. Documents move to Archive immediately.
Autonomous learning	Relevance model updates driven by Claude's own scores and Skipped stubs, without user feedback.
Baseline	A curated static JSON file encoding requirements, prohibitions, and key dates of a settled regulation. Always available, no API call needed.
Domain	AI Regulation or Data Privacy - the two regulatory domains ARIS monitors, each with separate vocabularies and scoring.
ELv2	Elastic License 2.0 - permits non-commercial use and modification; prohibits commercial sale or managed service offerings.
False-positive gate	The _NOT_AI_CONTEXTS blocklist in utils/search.py that blocks NAIC, MAID, PAID leave, and other non-AI acronyms.
Force Summarize	Bypasses the relevance pre-filter and sends all pending documents to Claude. Used when documents are Skipped unexpectedly.

Horizon item	A regulation that is planned, proposed, or advancing but not yet in force. Shown with anticipated dates and countdowns.
Posture score	A 0-100 compliance score from gap analysis. Higher is better.
RAG	Retrieval-Augmented Generation - the technique Ask ARIS uses to search the document and baseline index before asking Claude to answer.
Skipped	A document that failed the relevance pre-filter or that Claude rated below 0.35. Stored as a summary stub with reason visible in Documents view. Feeds the learning agent as a negative signal.
Source profile	A rolling quality score (0-1) per data source, updated by human feedback and autonomous learning. Domain-keyed.
Title-only validation	Relevance check against the bill title alone, not title + search keyword. Prevents LegiScan keyword results from auto-passing relevance.
UTF-8 encoding declaration	# -*- coding: utf-8 -*- on line 1 of a Python file. Required on Windows for any file containing non-ASCII characters.

ARIS - Automated Regulatory Intelligence System

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Built with Claude. Runs locally. Knows the rules.